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**St. Xavier's College of Education
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Effectiveness of Animated Games on Study Habits of Secondary School Students

Abstract

The present study belongs to Animated Games and Study Habits. Animated Games like visuals to create engaging worlds. Animated Games are not just for entertainment: many involve problem-solving, quick thinking, memory, and strategy, which stimulate cognitive skills. If gaming takes up much time or becomes a distraction, it can definitely hurt your study habits by eating into study time or making it harder to focus afterwards. Therefore, the objectives of the present study are to explore the Effectiveness of Animated Games on the study habits of secondary school students. This study hypothesises that there is no significant difference in the mean Study Habits Scores of the secondary school students before and after treatment of Animated Games. The Experimental research methods were adopted with a quantitative research approach. The study investigates the impact of animated games on the study habits of secondary school students. A Study Habits Scale was used in this present study, which was developed by Dimpal Rani & M. L. Jaidka. 20 students were selected through a purposive sampling technique from different colleges of Kanpur city, Uttar Pradesh. The data were analysed with the help of a t-test. The result of the study indicates that animated games had a significant positive effect on the study habits of secondary school students, as evidenced by the higher post-test mean scores ($M = 123.5$) compared to pre-test mean scores ($M = 110$). The findings of this study also reveal that there is a statistically significant difference in the mean study habits scores of secondary school students before and after exposure to animated games.

Keywords: AAnimated Games, Study Habits, Secondary School Students

Introduction

The more-than-half-century history of Animated Games reflects the evolution of arts, technology, and storytelling (**Pataranutaporn et al., 2021**). Animation in video games started with the experiments of early computer scientists in the mid-1900s. One of the first examples was Tennis for Two (1958), a game played on an oscilloscope that used simple graphical motions to simulate a Tennis match (**Pabon, 2016**). The subsequent Spacewar (1962) introduced basic animated movement, which established the groundwork for interactive gaming. The commercial video game industry started to take shape in the 1970s (**Squire, 2006**), propelled by arcade gaming. With Animated visuals, Atari's ground-breaking success, Pong (1972) imitated a table Tennis game. Other arcade classics, such as Space Invaders (1978) and Asteroids (1979), further employed animation to create dynamic and engrossing gameplay despite technological limitations (**Funk, 2021**). These games aided in the evolution of animation as a key element of video games. The 1980s were a golden period of gaming (**Maddison, 2001**) when home systems like the Sega Master System and Nintendo Entertainment System (NES) brought 8-bit graphics and lively animations to households. Characters like Link, Mario, and Sonic the Hedgehog became iconic thanks in part to vibrant (**Rossignol, 2009**), pixel-based animations. Side-scrolling games like Super Mario Bros (1985) showed how animated settings could enhance gameplay (**Bosman, & Van-Wieringen, 2022**). Some games from this era experimented with cinematic animation, including Dragon's Lair (1983), which used hand-drawn animated scenes to provide an interactive movie-like experience. Significant advances in 3D animation technology occurred in the 1990s (**Laine, 2018**), as completely 3D worlds were made possible by systems such as the PlayStation and Nintendo 64.

Realistic figures and three-dimensional settings were introduced by skeletal animation and PC games like Doom and Quake, which helped them become more and more popular. Motion capture technology revolutionised animation in the 2000s and 2010s (**Forte & Campana, 2016**) by enabling producers to produce lifelike character animations based on human movements (**Itti et al., 2004**). Stylised animation was employed in independent games such as Uncharted (2007) and The Last of Us (2013). Using cutting-edge technology like virtual reality, AI-driven animations, and ray tracing, modern animated games create immersive experiences (**Li et al., 2020**). From the pixelated simplicity of early games to the lifelike animations of modern

blockbusters, animated games have evolved into a powerful medium for narrative and visual expressions (**Jordan, 2009**). This long history illustrates how creativity and technology have come together to produce one of the most avant-garde forms of entertainment that is currently accessible: Animated Games.

Animation and games are two of the most popular forms of entertainment for children, and both significantly impact children's development. Children gain most of their early experiences through games (**Cramer, 2018**), requiring adult participation. Games on mobile devices that can be played by one or more players are referred to as animated games. These games are especially well-liked when they are available for free download (such as "freemium games," which are cost-free but require payment for additional features). Teenagers are more susceptible to internet-related addiction (**Selfhout et al., 2008**) than most other groups in society due to a combination of factors, including their developmental and psychological makeup, the ease with which they can access the internet through portable devices, and their positive expectations around mobile gaming. Addicts to animated video games have been shown to have worse mental and cognitive performance (**Ambuchi, 2014**), as well as more emotional problems, such as heightened anxiety and sadness and social isolation. Many studies have demonstrated the significant influence of games on cognitive processes (**Dang, 2021**).

All these technologies are very good at distracting people (**Cramer, 2018**). In line with this development, animated games were created to entertain people. According to Cush and Griffiths, teens who play animated games are simply having fun (**Dyer-Witthford, 2005**). They don't actually play because of any seriousness, but because they just want to feel relieved. When it was time for them to shape their future, they started showing their attention and interest towards animated games, which gave them both mental peace and happiness (**Eriksen, & Visentin, 2023**). A person can acquire skills, attitudes, and other behaviours that reflect practical values in the society (**Taylor, & Hamdy, 2013**) in which they live by reading the Dictionary of Education. Education is also defined as the social process through which individuals are exposed (**Brown, & et al., 1989**) to the influence of a chosen and controlled environment (particularly the school), in order to achieve social competence and optimal individual development. Teenagers' development of abstract and logical thought occurs during adolescence (**Kauer, & et al., 2012**). The person is prepared to take on responsibility and make decisions at this time.

Adolescence is crucial for the formation of a person's personality (**Ainiyah, & et al., 2024**) since it helps them strike a balance between their independence, autonomy, and responsibility (**Arnett, 1998**).

There is a claim that playing animated games provides numerous advantages., if not too many. A person may only be considered addicted to animated games if they are unable to control their need to play (**Van Holst, & et al., 2011**) and if this has a detrimental effect on their emotional, mental, and social lives. Animated games are classified as “drive control disorder,” which includes symptoms like not being able to control when to play animated games, not being interested in other activities, continuing to play animated games despite their negative effects (**Subrahmanyam et al., 2000**), and experiencing psychological exhaustion when you're not playing them. The internet and mobile phones are both a blessing as well as a curse to education (**Schwanen & Kwan, 2008**). It is considered to be a useful source of data that would enhance academic study and a threat to students' learning and academic performance. The overuse of smartphones to the point where it interferes with a person's everyday activities or life is known as smartphone addiction. It can also be viewed as an uncontrollable behaviour that interferes with day-to-day activities, causes significant stress and discomfort for friends and family, and negatively impacts an individual's academic environment.

In the present times, the importance of technology has increased so much (**Henriksen et Al., 2016**) that it has not only improved the learning of students but at the same time, some of its side effects are also seen. In the present times, students are becoming so influenced towards technology (**Al-Fuqaha et al., 2015**) that both positive and negative results are seen in them. A person with an addiction to animation games may also suffer from “gaming addiction” or “gaming disorder,” which is characterised by obsessive or uncontrollable behaviour linked to gaming. An individual's mental health, social relationships, education, and physical well-being are just a few of the positive and negative effects this addiction may have on a person's life (**Griffiths et al., 2012**). A fixation on gaming, withdrawal symptoms when one stops playing, increased tolerance (needing to play more to have the same degree of enjoyment), fruitless attempts to cut back or quit gaming, and neglecting one's obligations to one's personal and professional life are common symptoms and effects on Behaviour and Psychology. Even while addicted gamers are aware of the detrimental effects gaming has on their lives (**Serra et al., 2021**), they frequently have an overwhelming need to play.

A gaming addiction, especially if access to it is restricted or if it is used as an escape from real-life issues, can cause mood swings, anger, anxiety, and melancholy. Addicts may become disinterested in associating with friends and family, hobbies, and physical activities **(Busby, 1994)**. A sedentary lifestyle, which is exacerbated by prolonged gaming, can lead to obesity, bad posture, and related health problems like cardiovascular illnesses. Over gaming frequently results in sleep deprivation or irregular sleep cycles **(Pieters et al., 2014)**, which impairs general health and cognitive performance. Long periods of time spent in front of a screen can lead to headaches, carpal tunnel syndrome, and other repetitive strain problems **(Kumar et al., 2023)**. A gaming addiction can cause people to retreat from social obligations and in-person contacts, which can strain relationships and cause loneliness **(Garvey & Miller, 2021)**. Family members who are worried about the excessive gaming behaviour may have frequent arguments with you **(Jandriæ et al., 2020)**. Diminished Academic Achievement Playing video games too much might cause children to do poorly in class **(Xenos, 2009b)**, skip school, or even give up entirely since they can't concentrate on their studies.

As a result of research conducted in the second half of the century, adolescence has been accepted as the main stage of a person's life **(Kimberly, 2002)**. In this stage, many changes occur in students that have both positive and negative effects on their social and emotional intelligence. In the present times, the importance of technology has increased so much that it has not only improved the learning of students but at the same time, some of its side effects are also seen **(Friedman et al., 2000b)**. In the present times, students are becoming so influenced towards technology that both positive and negative results are seen in them **(Zimmerman, 2000)**.

Animated Games and Study Habits

The complicated and multidimensional topic of Animated Games' effects on Study Habits has attracted a lot of interest lately **(Michelson, 2017)**. Millions of people worldwide are captivated by Animated Games, which have become an essential component of contemporary entertainment **(Karmakar & Bhadra, 2023)**. Even though Animated Games are fun and help people unwind, they are also significant in how students concentrate, especially. Playing Animated Games can improve Study Habits in a number of ways **(Jordan, 2009b)**. For example, instructional animated games can boost student motivation and engagement by making learning enjoyable and participatory

(**Siricharoen, 2023**). Students are more likely to remember material and get a deeper comprehension of the subject matter when they are actively involved in the learning process. Furthermore, playing Animated Games might enhance cognitive abilities, including decision-making, critical thinking, and problem-solving (**Papoutsi & Drigas, 2016**). Playing games that call for planning and strategic thinking can help you develop these abilities, which are crucial for academic achievement (**Papanastasiou et al, 2017**). Additionally, Animated Games can foster imagination and creativity (**Calleja, 2010**), especially those that let users construct, design, or create virtual environments. Students get benefit from this, especially in courses like technology, design, and art. Additionally, playing Animated Games are providing children a sense of achievement and fulfilment, which can increase their self-esteem and confidence (**Chun, 2021**).

Playing Animated Games too much affects study habits (**Robson et al, 2015**). For instance, kids who play Animated Games for extended periods of time become distracted from their studies and perform worse academically. Students who play Animated Games for extended periods of time disregard their studies, homework, and other academic obligations (**Landoy, & et al., 2019; Coupaye, & et al., 2016**). Furthermore, excessive gaming is result in addition, which are harm to one's study habits and general well-being. A variety of issues, such as social isolation, a decline in physical activity, and poor academic achievement, result from a gaming addiction (**Thomée et al, 2010**).

Additionally, playing Animated Games late at night interferes with sleep cycles, which causes weariness and impairs concentration in class (**Miller, 2022**). Students who don't get enough sleep find it difficult to focus and remember things, which has a detrimental effect on their academic performance. Additionally, playing Animated Games too much causes social isolation (**Jordan, 2009c**), especially of adolescents spend more time playing than socialising with friends and family.

Finding a balance between studying and gaming is ultimately crucial to upholding effective study techniques and succeeding academically. Understanding the possible effects of animated games on study habits helps students prioritise their academic obligations (**Mills et al., 2022**) and make wise gaming decisions (**Coupaye et al., 2016**). Students can benefit from animated games while succeeding academically and upholding effective study habits (**Milburn, 2018**) if they take the proper approach.

Rationale of the Study

Animated games have become a popular tool in education, enhancing student engagement and motivation (**Garris et al, 2002**). However, their impact on study habits remains underexplored. Traditional study techniques often fail to sustain students' interest, especially in younger generations (**Branston, 2007**). Animated games, with their engaging narratives, interactive features, and immediate feedback mechanisms, have the potential to address these challenges (**Sailer et al., 2016b**).

This study aims to understand how animated games influence key aspects of study habits, such as attention span, time management, and self-discipline. It seeks to identify whether animated games can foster positive learning behaviours or contribute to distractions and poor academic performance (**Ciampa, 2012**). This research fills a significant gap in existing literature and provides educators, parents, and policymakers with evidence-based insights to make informed decisions regarding the use of animated games in educational settings.

Genesis of the Problem:

The emergence of Animated Games as a potential problem affecting study habits is a complex issue with both positive and negative impacts (**Westera, 2015**). On one hand, Animated Games can have cognitive benefits, such as enhancing problem-solving skills, critical thinking, and hand-eye coordination (**Patil et al., 2020**). They can also make learning more engaging and interactive, potentially increasing students' motivation to learn. However, excessive gaming can lead to negative consequences, including distraction, addiction, reduced attention span, sleep deprivation, and social isolation (**Aydin & Ku°, 2023**).

The widespread availability of gaming devices and internet access, targeted marketing and advertising, and lack of parental guidance and monitoring contribute to the problem (**Sarwatay et al, 2021**). To address this issue, parents, educators, and policymakers must work together to promote responsible gaming habits, set limits, and encourage a balance between gaming and academic activities (**Jackson & Games, 2015**). By doing so, they can help students maximise the benefits of animated games while minimising their negative impacts on Study Habits.

Objectives of the Study

- (1) To study the study habits of secondary school students before the treatment of animated games.
- (2) To study the study habits of secondary school students after treatment with animated games.
- (3) To compare the study habits of secondary school students before and after treatment with animated games.

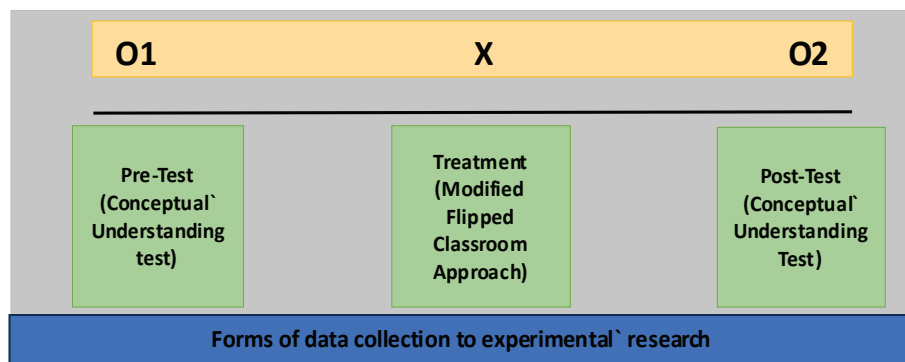
Hypothesis of the Study

- (1) There is no significant mean score difference between the Study Habits of the secondary school students at the pre-test and post-test levels.

Methodology

As per the nature of the study and its objectives, the researcher applied a One-Group Pre-Test Post-Test Design under the Pre-Experimental research method with the quantitative approach. This is an experimental type of research.

Figure-1: Research Design of the Study



The study was conducted on secondary school students. The sample is representative of the population, but in experimental research is limited to secondary school students. The population of the study comprised secondary school students (According to NEP 2020, classes 9th to 12th) of Kanpur city. Utilising a purposive sampling technique, the sample for the study was selected. 20 students (3 students from 9th class, 9 students from 10th class, 4 students

from 11th class & 4 students from 12th class) were selected in this study's final sample.

To assess the impact of Animated Games on the study habits of Secondary School Students A Study Habit Scale developed by Dimpal Rani & Dr M. L. Jaidka (2015) was used.

A paired sample t-test was conducted to assess the impact of animated games on the study habits of secondary school students before and after the intervention.

Analysis and Interpretation

The findings of the present study yield valuable insights into the impact of animated games on the study habits of secondary school students. The conclusions drawn from the study are discussed below:

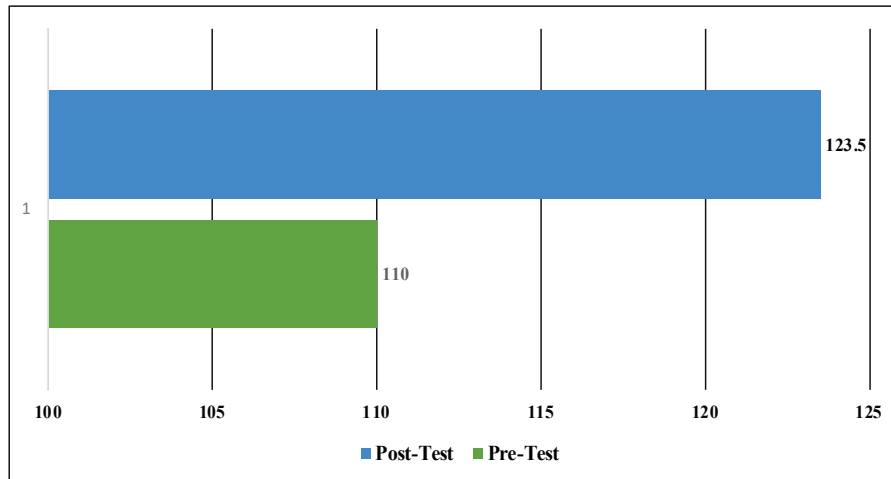
Hypothesis 1: There is no significant difference in the mean scores of study habits of secondary school students before and after treatment with animated games.

Table-1: Depicts the Effectiveness of Animated Games on the Study Habits of Secondary School Students about the mean difference of pre-test and post-test scores

Variable	Test	N	Mean	S.D.	S _D	df	t-value	Result
Study Habits	Pre-Test	20	110	11.22	3.99	19	3.38	Significant at the 0.01 level
	Post-Test	20	123.5	13.34				

The t-value 3.38 indicates that there is a significant difference in the mean scores of study habits of secondary school students before and after treatment with animated games. *The study found that students' study habits were significantly higher after they played animated games compared to before. Specifically, the mean score for study habits after the treatment (playing animated games) was higher than the mean score before the treatment. This indicates that playing animated games had a positive effect on the students' study habits.*

Graph 1 illustrates this difference, showing a clear increase in the average study habits scores of secondary school students from the pre-test (before exposure) to the post-test (after exposure to animated games).

Graph-1: Mean difference of Secondary School Students

Discussion

The result of the current investigation corroborates those of Liu & et al., (2024), Yang and Gong, (2024), Kamal and Wok, (2020), Leila and Lila, (2020), Kuo-Ming Chu, (2023), Rayes & et al, (2023), Chatterjee & et al, (2022) regarding the significance positive impact of serious games Liu & et al., (2024) on body composition in children and adolescents. Likewise, results of all research works like Olipas & et al (2020), Fikri (2024) proved that playing online games has both positive and negative effects Fikri, (2024) on students' school achievement in English lessons. The research works of Vashist & et al., (2024), were found consistent with the finding that majority of the adolescents had positive attitudes of using BGMI games in the same way Rayes & et al., (2023), in their research results observed that the relationship between the level of mobile dependency and socialization of high school students were found to be significant and positive while Chatterjee & et al., (2022), observed utilitarian attitudes of people towards playing mobile games with different factors. Meddage et al., (2024), Kim et al, (2024), Irmayani and Anas, (2020), Liu and Li, (2011), Khoirunnisa and Ramadan, (2024) were found their research results that the positive impact of online games on the school behaviours while Mun, 2022, Rogers, (2016), observed significant and positive relationship between online gaming addiction and mental health. Olipas (2020) found that mobile games have a moderate positive impact and minor negative impact, as did Connolly & et al (2012), focused on the positive impacts of playing games. Khoirunnisa and Ramadan (2024) found in their research results

that the psychological impact of playing the online free-fire game for children aged 8 years was found to increase the child's ability to concentration train problem-solving skills.

In more research studies like Meddage (2024), Rayes & et al (2024), explored a complex relationship between video gaming habits and sleep patterns as well as Khoirunnisa and Ramadan (2024), who observed unstable emotions, thinking disorder, losing concentration and behavioural disorder. Syahrul et al, (2024), were found that negative impacts of online game addiction on nine dimensions: Moral, Educational, Health, Psychological, Social, Physical, Financial, Character, and Spiritual while Hamila and Rustan, (2023), observed the free-fire online game addiction impact on emotional changes, interest in learning and lack of interaction as well as Kuo-Ming Chu, (2023), found mobile game addiction was negatively connected with academic performance and mobile self-efficacy. Haberland and Atkin (2022) observed in their study that mobile game addiction shows a negative relationship between social support and internet addiction, while Husna et al (2022) proved that online gaming addiction is negatively associated with academic performance. Seok and DaCosta, (2015, Hamila and Rustan, (20230, Sayeed & et al, (2021), observed in their study that the majority of the students played mobile legends and spend mostly hour playing mobile games while Sayeed & et al., (2021), were found the uses of mobile games suffering from several physical problems such as: Headache, Eye-Discomfort, Blurry-Vision, and Ear-Discomfort while Wang & et al, (2019), were found more social anxiety and Rogers, (2016), was found to be mental patients of using mobile games as well as Gao & et al., (2014), proved that games affects the attitudes Lin & et al., (2011), towards intrinsic and extrinsic motivations.

Conclusion

The study presents some exhaustive conclusions regarding study habits. The variables were studied against the impact of animated games on the study habits of the secondary-level students of Kanpur city, who were selected for the research study.

The study revealed that the study habits of secondary school students improved significantly after exposure to animated games. The post-test mean score ($M = 123.5$) was notably higher than the pre-test mean score ($M = 110$), indicating that animated games had a positive effect on the students' study habits rather than hindering them.

While the findings suggest that animated games may contribute positively to students' academic routines, the researcher acknowledges that the improvement could also be attributed to factors such as test familiarity, practice effect, or heightened motivation due to the novelty of the intervention. Therefore, the results should be interpreted with caution.

Nevertheless, this study provides preliminary evidence that animated games, when used appropriately, may serve as a useful tool to enhance student engagement and study habits rather than merely as a source of distraction. It is recommended that educators and parents encourage balanced usage of animated games, integrating them as supplementary learning aids while ensuring that students do not develop excessive dependence on them.

Further research with larger sample sizes, longer intervention periods, and controlled experimental designs is recommended to establish the long-term effects of animated games on study habits and academic performance.

A significant difference was observed in the study habits of students before and after playing animated games. Specifically, their study habits were considerably better before exposure to the games. Importantly, gender was not found to be a determining factor in predicting study habits among the participants.

Based on these findings, it is recommended that students be educated about the risks associated with excessive use of animated games. Such awareness can help them understand the negative consequences on their academic routines and encourage them to use these games only for relaxation and enjoyment, rather than as a habitual escape. This approach would also reinforce the importance of developing effective and consistent study habits.

In the current educational system, which is largely exam-oriented and heavily reliant on mental performance, students are experiencing considerable physical and psychological unrest. Many find themselves in constant conflict with themselves, their parents, and their teachers. This has led to rising levels of tension, frustration, and unnecessary anxiety. A lack of self-awareness and concentration is pushing students into a cycle of stress, where they often turn to animated games as a coping mechanism.

Therefore, this study serves as a valuable contribution by highlighting the factors—particularly animated games that adversely affect students’ study habits and academic well-being.

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